Main Color Models

Main Color Models

- **Additive Color Model**
  - Colors perceived are the result of **transmitted** light
- **Subtractive Color Model**
  - Colors perceived are the result of **reflected** light

Additive Color Model

- For computer and TV displays
- Uses light to display color
- Colors result from transmitted light
- Red + Green + Blue = **White**
- Known as the **RGB** color model

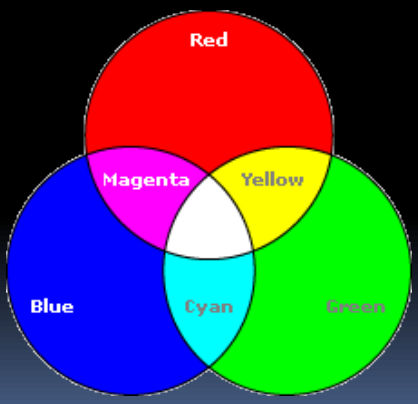
Subtractive Color Model

- For printers
- Uses ink to display color
- Colors result from reflected light
- Cyan + Magenta + Yellow = **Black**
- Known as the **CMYK** color model

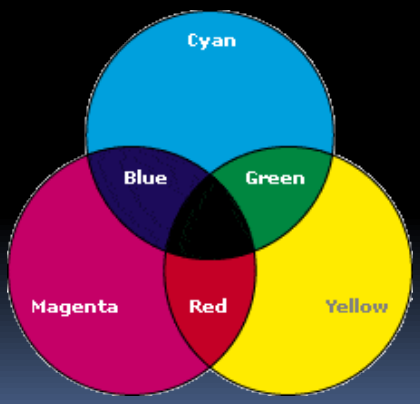
RGB vs. CMYK



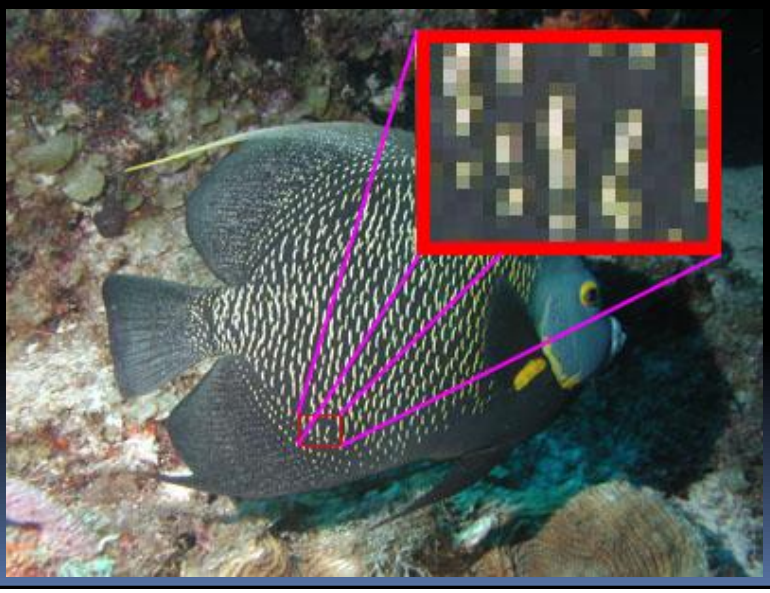
Additive Model RGB



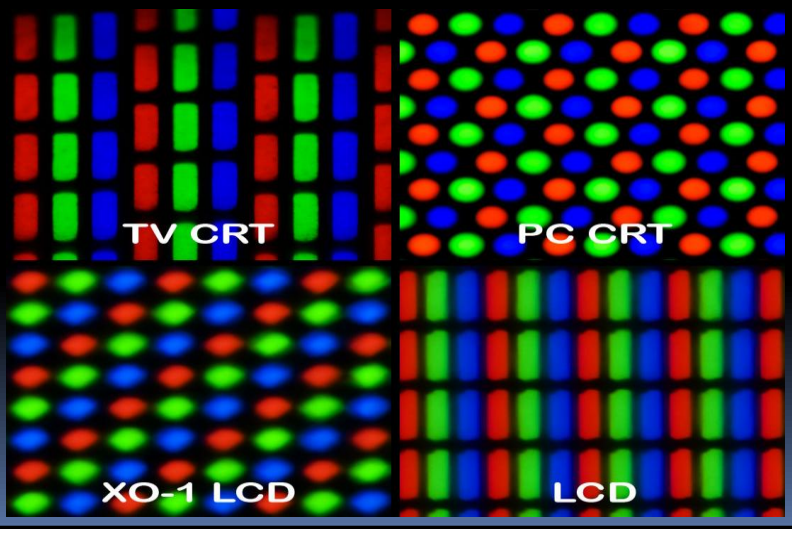
Subtractive Model CMYK



Each image consists of pixels



Colors are arranged differently in different monitors



Color Basics

Color Basics

- A color is described in three ways: by its name, how pure or unsaturated it is, and its value or lightness.
- Although pink, crimson, and brick are all variations of the color red, each hue (color) is distinct and differentiated by its intensity and value.

Formal Basic Definitions

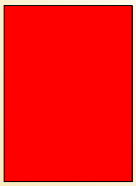
- **Hue** or **Color**: An angle on a color wheel.
- **Chroma / Saturation / Intensity** : The degree of purity of a hue. The brightness or dullness of a hue. May be obtained by adding gray to the hue.
- **Luminance / Value**: A measure of the amount of light reflected from a hue. Those hues with a high content of white have a higher luminance or value.
- **Shade & tint**: Terms that refer to *value* variation of a hue.
  - **Shade**: A hue produced by the addition of black.
  - **Tint**: A hue produced by the addition of white.

Main Color Qualities

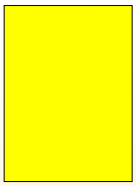
- **Hue**
- **Value (Tint and Shade)**
- **Intensity (Saturation)**

HUE

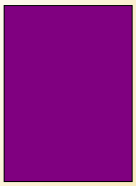
The name given to a color.



RED



YELLOW



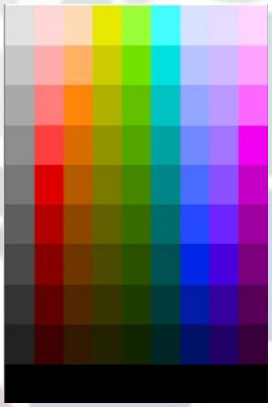
VIOLET

Value



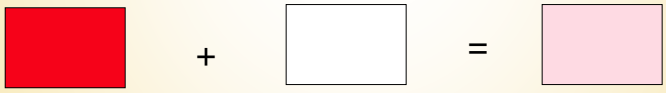
Here is an example of a value scale that has values ranging from the darkest dark, to the whitest white.

- Value is the lightness or darkness of a color.
- You can obtain different values by adding black or white to a color.
- A light color is called a tint of the original hue. For example, pink is a tint of red.
- To make a color darker in value, black is added. A dark color is called a shade of the original hue. Maroon is a shade of red.



TINT

Made by adding white to a color so that it is lighter.



HUE + **WHITE** = **TINT**

SHADE

Made by adding black to a color so that it is darker.



HUE + **BLACK** = **SHADE**

COLOR MIXING

Tints and Shades



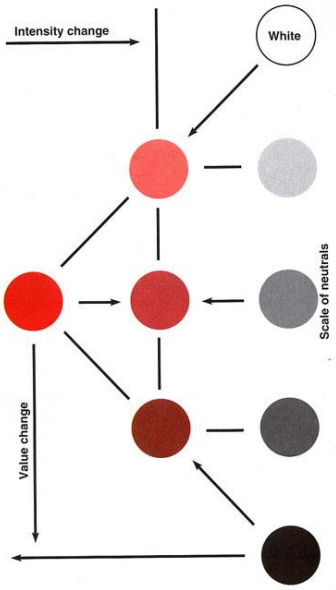


Fields in a Rising Storm

By Vincent Van Gogh

•This painting by Vincent Van Gogh, *Fields in a Rising Storm,* has tints and shades of blue in the sky, and tints and shades of green in the fields.

INTENSITY



- Intensity refers to the brightness or dullness of a color. An example is bright red (or dull red).
- When a hue is strong and bright, it is said to be high in intensity.
- When a color is faint, dull, and gray, it is said to be low in intensity.



Intensities of Green

NEUTRALS

(NOT CONSIDERED AS COLORS)

White

Black

No color or all colors

Gray

White + Black

Beige

Brown

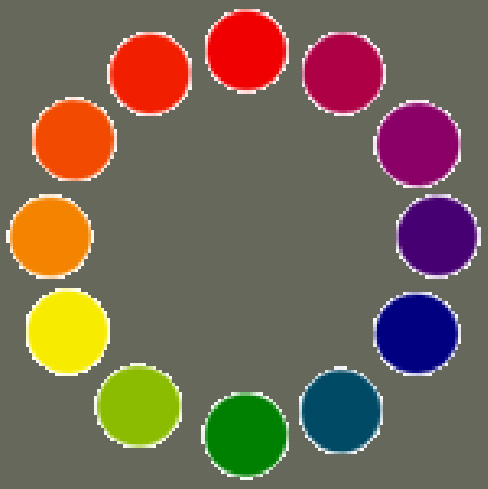
Can be used with most colors

Color Wheel

Color Wheel

- A color wheel (also referred to as a color circle) is a visual representation of colors arranged according to their chromatic relationship
- Begin a color wheel by positioning primary hues equidistant from one another, then create a bridge between primaries using secondary and tertiary colors

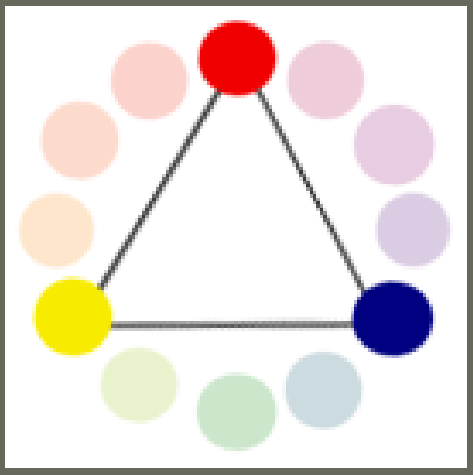
Color Wheel Example



Color Wheel – Primary Colors

- Colors at their basic essence
- Colors that cannot be created by mixing others
- Equal distance from each other on the color wheel
- All other colors are made from these

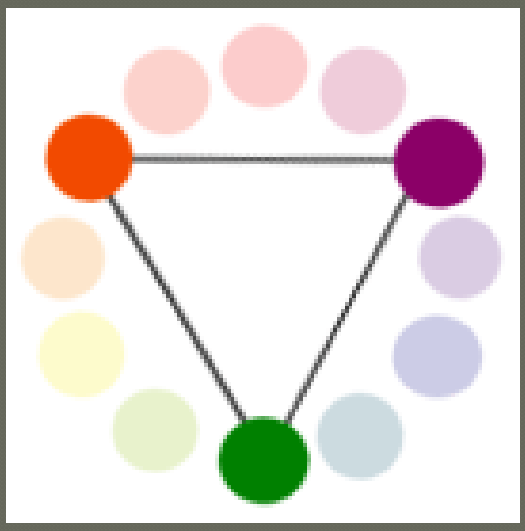
Color Wheel – Primary Colors



Color Wheel – Secondary Colors

- Colors achieved by a mixture of equal amounts of two primary colors
- Found halfway between the primary hues on the wheel

Color Wheel – Secondary Colors



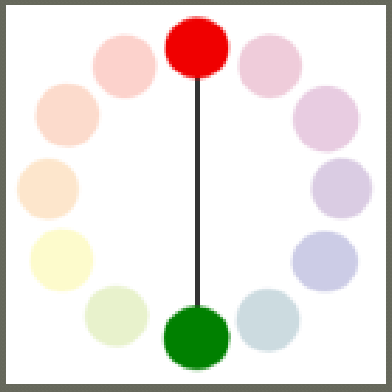
Color Wheel – Tertiary Colors

Colors achieved by a mixture of equal amounts of adjoining primary and secondary hues



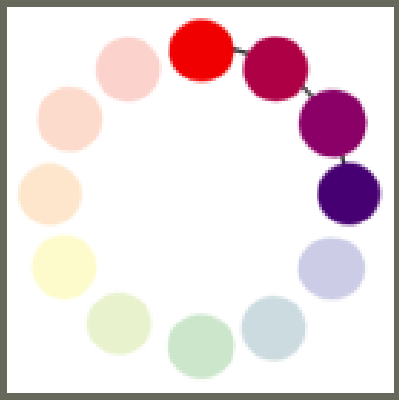
Color Wheel – Complementary Colors

Colors located opposite each other on a color wheel



Color Wheel – Analogous Colors

Colors located close together on a color wheel



Color Triangle

Color relationships may be displayed as a color triangle instead of a color wheel

